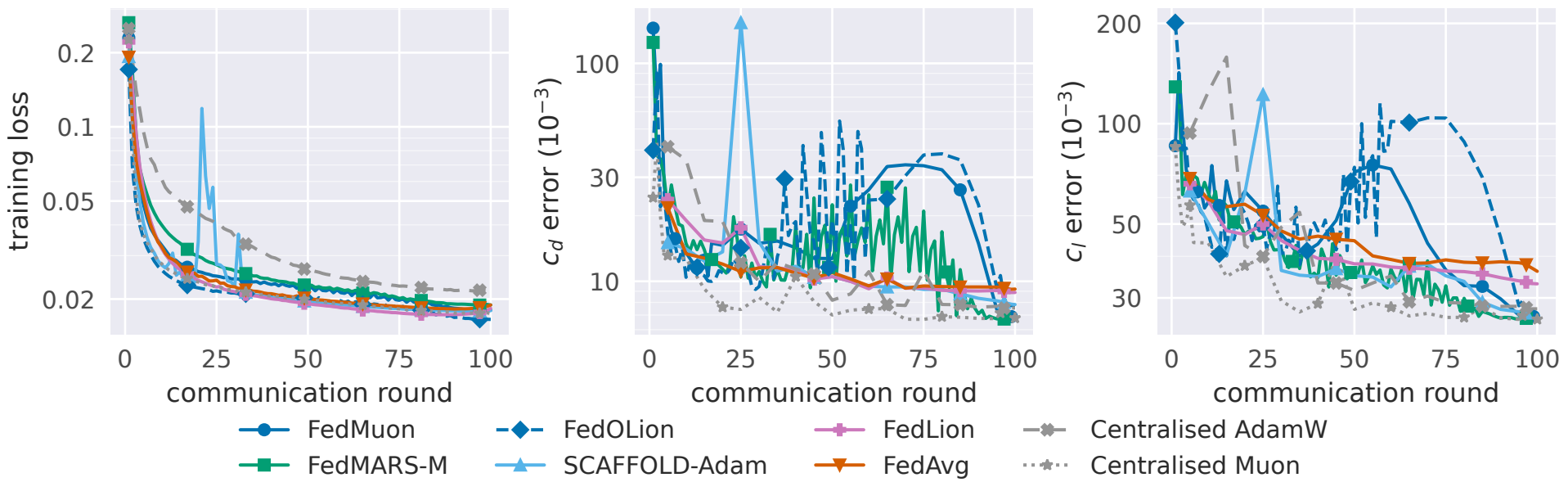
training loss
0.2
0.1
0.05
0.02
0
25
50
75
100
communication round
c_d error (10^{-3})
100
30
10
0
25
50
75
100
communication round
c_l error (10^{-3})
200
100
50
30
0
25
50
75
100
communication round
FedMuon
FedMARS-M
FedOLion
SCAFFOLD-Adam
FedLion
FedAvg
Centralised AdamW
Centralised Muon